THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

Name: Vinit · B · Lunia

Class: TI Roll No: 63

ASSIGNMENT NO: 1

Q1:- Define a project. What is project management? Explain various characteristic of project management.

Ans:- • A project is a temporary endeavour undertaken to create a unique project or service. Unlike "operations", a project has a specific beginning and an end. Its completed when its goal are met or when project is terminated.

↳ One-time (non-routine)
↳ Limited funds/time (one phase or multiple phase)
↳ Performed by people - single or multi people.
↳ Planned & controlled

• Project Management is the application of knowledge, skills, tools & techniques to project activities to meet the project requirement. It involves Planning, Organizing, Staffing, Directing, Monitoring, Controlling, Innovating & Representing.

The 4 P's of Project Management are

1. The people :-
2. Product
3. Process
4. Project

• Characteristics :-

1. Temporary Nature :- every project has defined start & end date. Its not permanent functional unit. Once the objective is achieved, the team is usually disbanded or moved.

2. Unique deliverables :- the outcome of a project is unique. Even if two bridges look the same, their location, environment or stakeholder will differ.

3. Progressive Elaboration :- project often starts with a broad idea & are refined in detail as project progresses.

4. Cross-Functional teams :- project management involves people from different departments coming together from to work toward a common goal.

5. Goal Oriented :- everything in project management is driven by set of objectives.

6. Risk & Uncertainity :- because projects involve doing something new or unique, they carry a level of risk. Project management involves identifying these risks early & creating mitigation plans.

Q2 : Write short note on PMBOK.

Ans: • Project Management Body of Knowledge Guide (PMBOK) prepared by PMI, USA, includes proven traditional as well as innovative practices in the area of project management. The purpose of it is to identify the "good practices" that are applicable to most projects, most of time & there is wide spread consensus about their value & usefullness.

• PMBOK proves useful for

→ Senior executives
→ Project Managers
→ Functional Managers
→ Project Team Managers
→ Stakeholders
→ Trainers
→ Consultant
→ Researchers

• Knowledge Areas :-

1. Project Integration Management KA :- includes the processes & activities needed to identify, define, combine, unify & coordinate various processes.

2. Project Scope Management :- it assures that project work expected from the team is defined accurately & correctly and that work is undertaken & completed as well.

3. Project Time Management :- it identifies project phases & activities & then estimates and sequences them.

4. Project Cost Management :- includes the processes involved in planning, estimating, buggeting & controlling costs as that the project can be completed within approved budget.

5. Project Quality Management :- includes all activities that determine qualities policies, objectives & responsibilities so that the project will satisfy or exceed the need of stakeholder.

6. Project Human Resources :- focuses on managing most important resource of project. Includes process that organise & manage project teams.

7. Project Communication Management :- focuses on developing processes required to ensure timely, accurate & appropriate generation, collection & distribution of project information.

8. Project Risk Management :- includes all processes associated with risk management planning, identification, response & monitoring on a project.

9. Project Procurement Management :-

Q3: Explain ITPM.

Ans: • Information Technology Project Management (ITPM), is specially tailored to managing projects that involve hardware software, network & data. IT usually deals with intangible products, rapidly changing technology & high levels of integration with existing business systems.

• Core Components :-

1. IT Project Lifecycle (ITPLC) :- unlike standard, IT project are function a specific software or system

development methodology.

→ SDLC :- Software development life cycle is structured process including requirement analysis, design, coding, testing, & maintenance.

→ Agile / Scrum :- iterative approach where project is broken into small "sprints" to allow frequent changes

2. Integration Management :- IT projects rarely exist in vaccum. ITPM ensures that a new software systems can "talk" without crashing the entire infrastructure

• Key Challenges in ITPM :-

→ High Obsolescence :- tech evolves so fast that a tool might be outdated by the time product launches.

→ Intangibility :- its harder to visualize progress in 10000 line of codes than its to see physical progress.

→ Security & Privacy :- every IT project must account for data protection, encryption & cyber threats.

→ User Resistance :- implementing new IT system often requires changing how employees work.

• Importance :-

→ Reduce Risks :- through better testing & prototyping

→ Improve ROI :- by ensuring technology actually solves a business problem.

→ Enhance quality :- by utilizing standardized code & deployment practices.

Q4: What is role of RFP ? Describe its content & purpose.

Ans:- • A Request For Proposal (RFP) is a document that's used to solicit bids, quotes or proposals for prospective suppliers. Its an early stage process document, issuing an invitation for suppliers, often through bidding purpose.

• The idea behind an RFP is to prepare a level playing field for all perspective suppliers by stating in clear terms the requirements of the organization & specification.

• Purpose :-

→ Standardization :- ensures all vendor receive same information.

→ Problem Solving :- an RFP defines a problem & asks vendor to purpose their unique solution.

→ Risk Mitigation :- by clearly refining technical requirement & legal terms upfront.

→ Objectivity :- provides structured framework for evaluating vendors.

• Process :-

PROJECT PLAN → DRAFT RFP → ISSUE RFP → REVIEW PROPOSAL → AWARD CONTRACT

• Content :-

1. Introduction & Background :- Give purpose of RFP, project objective & additional background information about project or business objective

2. Administrative Information:- Technical contact, Contractual contant or due dates.

3. Guidelines for Proposal Preparation:- Proposal submission, award contract which is by vendor is more advantageous

4. Evaluation factors for awards:- completion of all required responses, vendors proposed solution meets all requirement vendor's ability to deliver, overall cost & experience

5. Scope of Work:- list specific task, technical requirement & functional specification.

6. Deliverables:- approach, findings, recommentations associated with project.

Q5:- What is feasibility study? Explain its type, content & purpose.

Ans:- • A Feasibility Study is a preliminary study of a project's various elements to determine its potential for success It describes how easy or difficult its to accomplish a set goal & is used to identify obstacles & form strategies.

• Purpose:- primary goal of the study is to "get it right" before investing resources, time or money.

1. Decision Support:- providing pertinent information to aid in making a "go / no-go" decision on whether to proceed.

2. Risk Identification:- highlighting potential issues & hurdles that could lead to project failure

3. Success Assessment :- measure the project's ability & likelihood of being completed effectively

4. Alternative discovery :- finding new discover or narrowing down business alternative.

• Types of Feasibility Study :-

1. Technical feasibility :- focuses on the technical resources available to the organization & whether technical team is capable of converting ideas to working system.

2. Economic feasibility :- involves a cost-benefit analysis to identify how well or poorly a project will conclude financially

3. Legal feasibility :- investigates if any aspect of proposed plan conflicts with legal & ethical requirement.

4. Operational feasibility :- analyses how well the proposed system solves problems & meets the organization's need.

5. Scheduling feasibility :- a measure of how reasonable the project duration is & if it can be completed on time

• Contents of a feasibility studio :-

1. Introductory analysis :- summarise project concept, market circumstances & potential hurdles.

2. Scope definition :- evaluates the project's influence on internal stakeholder & external customers.

3. Financial statement :- contains projection income statement and an "opening day balance sheet" to show capital, asset

4. Market Research :- includes population trends, demographics & market volume to establish revenue projections

5. Roadblock & Alternatives :- research into measure to assure success if barriers develop during investigation

6. Final decision :- clear recommendation on viability.

Q6: Write a short note on W.B.S ?

Ans: • The Work Breakdown Structure (WBS) is a logical decomposition of the work to be performed in a project. It serves as an outline that focuses on how the project, service or result is naturally subdivided into manageable parts

• Key concepts & components :-

1. Work package :- the lowest level of the WBS, representing the actual task to be executed.
2. Deliverables :- tangible, verifiable work products such as reports, presentations or prototype.
3. Milestones :- significant events or achievements, such as the acceptance of a phase or a "Crux".
4. PMBOK alignment :- WBS is the foundation for activity definition, sequencing, duration estimation & schedule developmen

• How to create a WBS (6-step process)

1. Collecting project requirement

2. Establishing significant project phase & deliverable
3. Breaking down proses into smaller tasks
4. Assigning time estimate to each task
5. Verifying tasks are unique
6. Gathering feedback from the team.

• Types of WBS :-
1. Task-oriented :- defines the project in terms of steps, phases or functions.
2. Deliverables oriented :- focuses on the tangible physical components needed.
3. Process oriented :- focuses on the processes involved in the project life cycle.
4. Hybrid :- combination of deliverable & task oriented structure
5. Time-phased :- used for long projects, breaking them into major phases instead of specific tasks.

Q7:- Consider a XYZ company under take a project to computerize working of ABC City Bank, Then develop W.B.S.

Ans:- To computerise the bank, we follow Work Package Concept to decompose project into specific, measurable deliverable

• Level 1 :- Project Management & Planning
→ Requirement Collection :- interviewing stakeholder to define banking functions.
→ Feasibility :- study assesing technical, economic & legal viability
→ WBS Finalization :- detailed task breakdown & team feedback.

- Level 2 : Infrastructure
  - → Hardware procurement :- servers, teller terminals & networking equipments
  - → Network Installation :- setting up secure LAN/WAN for bank branches
  - → Security setup :- firewalls & physical security for server

- Level 3 :- Software Systems
  - → Database development: customer records, transactions history and account data.
  - → Core Banking modules: software for deposit, withdrawal & interest calculation.
  - → User Interface :- developing dashboard for tellers & administrator.

- Level 4 :- Testing & Quality control
  - → Crux : verifying the core transaction engine works
  - → Integration Testing :- ensuring the software communicates correctly with hardware.
  - → User Acceptance Testing: bank staff testing the systems for usability

- Level 5 :- Implementation & close out
  - → Data migration: digitizing manual bank records into the new system
  - → Staff training :- testing bank employees on new digital processes.
  - → Project handover :- providing final documentation & recommendation.